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SCD: Segment-CNN-Decision for Signal Detection in Wideband Spectrum with Unknown Interference

Tianyi Zhao, Connor Steigerwald*, Johnny Zhang* and Danijela Cabric
Electrical and Computer Engineering Department, University of California, Los Angeles, USA

Email: zhaotianyi@ucla.edu, csteigerwald1@ucla.edu, jzhang566@g.ucla.edu, danijela@ee.ucla.edu

***Abstract*—Specific signal detection in a wideband spectrum is important to ensure efficient and secure spectrum sharing. However, the signals can be subject to different signal-to-noise ratios (SNRs), varying channel conditions and even coexist with unknown interferences. In this work, we propose the Segment-CNN-Decision (SCD) framework, which is developed to detect 5G-like signals from raw I/Q data using real spectrum data collected over-the-air (OTA). SCD takes raw I/Q samples of a 1-second capture as input and crops it into multiple segments. Then, each segment is processed and downsampled to extract frequency-domain features. All features are sent to a concatenated convolutional neural network (CNN) model for initial detections. Finally, the separate detection results from all segments gathered by the decision module in SCD for the final detection result. This final detection result is a binary output indicating the existence of an target user transmissions in the input capture. Our proposed framework achieves 98% accuracy across all available training captures and 89% on a hidden testset.**

***Index Terms*—Wideband spectrum sensing, signal detection, machine learning.**

I. Introduction

Dynamic spectrum sharing (DSS) is essential in many wireless network such as 6G technology [1]–[3] and the Citizens Broadband Radio Service (CBRS) band [4]. To facilitate sharing and coexistence between heterogeneous wireless users, signal detection and classification techniques are important to protect incumbent users and ensure wireless security [5]–[7].

Many works have investigated the signal detection and classification problem from different aspects. Typical signal classification methods usually consider narrrowband signals which have been synchronized and extracted from received captures [8]–[11]. On the other hand, signal detection in a wideband spectrum generally aims to detect any signals existing in a designated band [12]–[14]. The problem of detecting target incumbent users with interferences is considered in [6], [7], where the interference signals are assumed to be controlled with known parameters.

In this work, we consider the wideband spectrum signal detection problem, in a more realistic and versatile environment, where many non-ideal effects can exist in the wideband captures, including varying SNRs, unknown interference signals, and varying channel conditions. To address this problem, we utilize the convolutional neural network (CNN) and propose the segment-CNN-decision (SCD) framework. Our contributions can be summarized as follows:

- We propose a machine learning based SCD framework which can be used for specific user signal detection in wideband spectrum. We show that the framework can be developed with very limited amount of training data and is robust against unknown interference signals and realistic channel conditions.
- We develop and evaluate the SCD framework on a real spectrum field dataset collected OTA [15]. The dataset includes unknown interference signals other than the target transmissions, a wide variety of SNRs and many different channel conditions, both stationary and mobile. The SCD framework is trained with only 240 captures and able to effectively detect the target transmissions with 89% accuracy in a hidden test dataset.
- We analyze the implementation efficiency of the SCD framework. Specifically, we discuss how different implementation methods can enable tradeoffs between space and computational complexities to be adapted to different needs and constraints.

The rest of the paper is organized as follows: Section II formulates and explains the problem. Section III describes the dataset used for developing and evaluating the proposed SCD framework. Then, the pre-processing and implementation details are discussed in Section IV. The evaluations and experiments are presented in Section V. Finally, Section VI concludes the paper.

II. Problem Statement

In this work, we consider the problem of detecting the existence of active user transmissions of 5G-like signals in wideband captures under realistic conditions and impairments. Specifically, the target 5G-like transmissions experience diverse channels, both stationary and mobile, and appear across a wide range of signal-to-noise ratios (SNRs). Meanwhile, the wideband captures also contain various unknown interference signals due to radio astronomy facilities in the 7.125 to 7.4 GHz band [15].

As shown in Figure 1, each capture may or may not contain the target user transmissions, together with different unknown interference signals. The detection objective is to determine whether a target active user transmission exists in any given capture. Therefore, the problem is formulated as a binary

*These authors contributed equally to this work.

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classification where 1 represents “with active transmitters” and 0 represents “without active transmitters”.

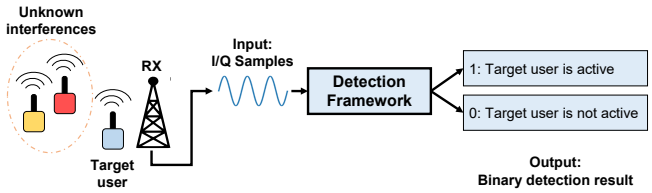


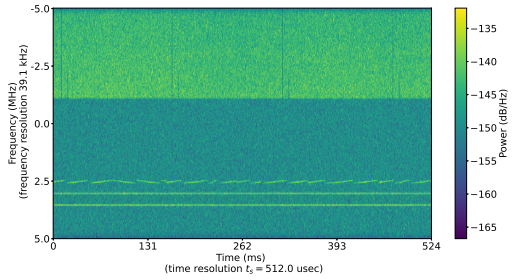
Fig. 1: SpX DAC problem statement.

This problem is challenging because there is only minimum prior knowledge about both the interference signals and the target 5G-like transmissions. Specifically, there is no prior information about parameters such as subcarrier spacing or preambles for the target transmissions. An example of capture segments with active target transmissions are shown in Figure 2. From Figure 2(a), we can observe that the target transmission exists in the lower half band of the captures with some visible low-frequency patterns, while in Figure 2(b) we can see there are some strong narrowband interference signals coexisting in the lower half band.

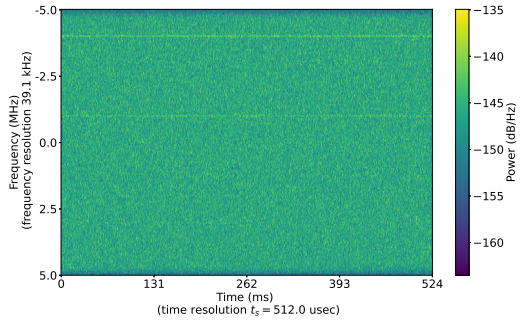
III. Dataset

To develop our framework, we utilize the a subset of the dataset collected during NSF SpectrumX’s July 2025 VLA experiment [15]. The available dataset subset consists of 240 1-second labeled captures of a 10 MHz bandwidth spectrum data from 7.125 to 7.135 GHz, with 10 million complex-valued I/Q samples. There are 160 positive captures which include the target user transmissions and 80 negative captures which may contain unknown interference signals. The target user transmissions are transmitted from 3 different locations with unknown transmission parameters, and the captures are collected from 5 different receive locations, including stationary and mobile scenarios [16].

To better utilize the training data, we first randomly select 64% captures as training dataset, 16% as validation set and 20% as test set (64-16-20 split). The training dataset is used for training the CNN in our proposed framework, which is discussed in details in Section IV. First, we repeat the training process several times with different randomizations in dataset split. Then, based on the evaluation results across different randomizations, we find 20 positive captures (with target transmissions) and 13 negative captures (without target transmissions) that are often incorrectly classified. With this observation, we group these 33 captures into a “hard” category, and the remaining 207 (140 positive and 67 negative) captures are thus grouped into an “easy” category. Finally, we apply the random 64-16-20 split again on both easy and hard captures for separate training, validation and test sets. It should be noted that there is a large-scale hidden test dataset in addition to the 240 available training captures. This hidden dataset will be used to evaluate the developed framework at the end. Further processing of the signals and framework development details will be discussed in Section IV.



(a)



(b)

Fig. 2: Example of captures with active transmissions under (a) high SNR and (b) low SNR scenarios.

IV. Proposed Approach: SCD

The overall workflow of our proposed Segment-CNN-Decision (SCD) framework is presented in Figure 3. As shown in Figure 3, the raw I/Q samples of a 1-second capture is first sent to the pre-processing block, where the capture is converted to N feature vectors of shape $W \times 2$. The feature vectors are then sent to a prediction module, which will output N values within the range $[0, 1)$. Here, each value could be interpreted as the likelihood that an active target user exists in its corresponding segment. Therefore, we can obtain N predicted likelihoods from the prediction module for each capture. These likelihood scores are then sent to a decision module, which performs thresholding and voting on the likelihoods and outputs a final detection decision. The details of the pre-processing pipeline, prediction module and decision module in the inference stage are discussed in Sections IV-A, IV-B and IV-C, respectively. The detailed implementation to develop SCD is explained in Section IV-D. Finally, we analyze the computational complexity of SCD in Section IV-E.

A. Pre-processing

First, we divide each capture into N segments of T samples, where every two consecutive segments have a $\rho\%$ overlap. Segmenting the captures with overlap is important to both the development of SCD and the detection performance in the inference stage. It augments the training data by increasing the amount of training samples, and also improves framework precision by allowing tolerance of missed detected segments.

Then, we compute the fast Fourier transform (FFT) of each segment, and take the absolute values of the results. We further

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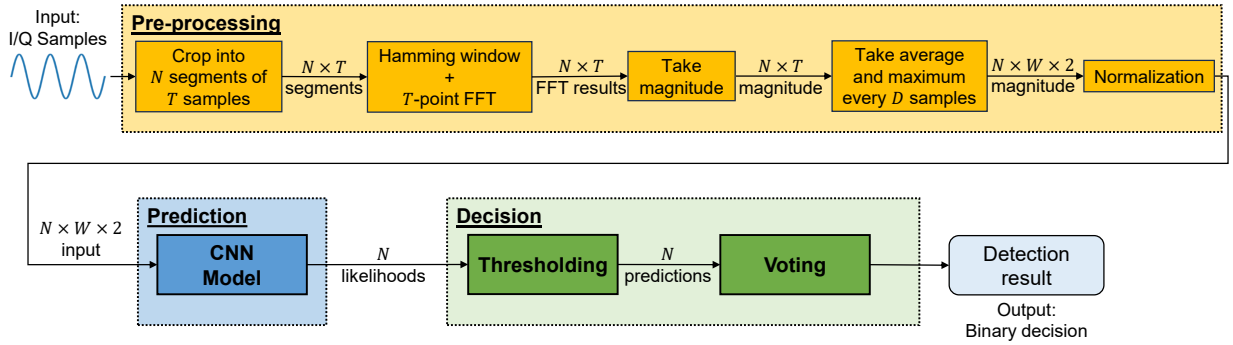


Fig. 3: Workflow of the proposed SCD framework.

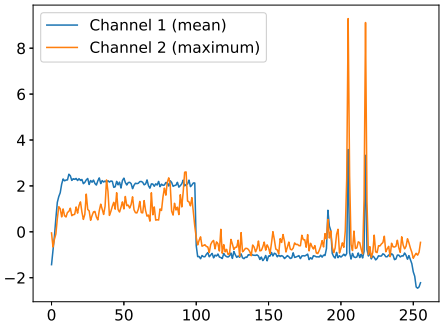


Fig. 4: Example of a positive normalized feature plot.

perform a downsampling of the FFT results by taking the average together with the maximum every D samples, resulting in $N \times W \times 2$ samples where $W = T/D$. Specifically, we interpret this result as N segments of W frequency-domain features with 2 channels for each capture. Finally, we normalize the results following Equation (1), where $i \in \{1, 2, \ldots, W\}$ corresponds to the $i-$th feature and $j \in \{1, 2\}$ corresponds to the $j-$th input channel. Moreover, μ_j and σ_j are the mean and standard deviation of the $j-$th channel across the training data. The overall pre-processing pipeline is shown in Figure 3.

$$\tilde{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j} \tag{1}$$

To select the hyperparameters, we start with visualizing a few captures. In Figure 2(a), we can observe that the transmission in the lower-half band is cut off approximately every 100 ms. Therefore, we select $T = 2^{20} = 1048576$ and $\rho\% = 80\%$ so that each segment can encapsulate enough information while allowing variances across different segments. Once T and ρ are selected, $N = 43$ is fixed in a capture of 10 million samples. Then, we select $W = 256$ to minimize irrelevant noises while preserving enough information in the downsampled frequency-domain representations. With T and W selected, we obtain $D = 4096$. Therefore, each segment is downsampled to a frequency representation of shape 256×2, so that each 1-second capture is converted to $43 \times 256 \times 2$ normalized feature vectors after the pre-processing stage. An example of a single pre-processed segment is shown in Figure 4.

TABLE I: CNN architecture.

Input size	Layer	Activation
256×2	Conv1D (32, 9, 3)	SiLU
83×32	BatchNormalization	-
83×32	Conv1D (64, 9, 3)	SiLU
25×64	BatchNormalization	-
25×64	Conv1D (128, 9, 3)	SiLU
6×128	GlobalAveragePooling	-
128	BatchNormalization	-
128	Dropout (0.5)	-
128	Dense (1)	Sigmoid

B. Prediction Module

The prediction module consists of convolutional neural network (CNN) which takes the pre-processed feature vectors as input. Therefore, the input shape to the CNN model is 256×2. Because we have a limited number of training samples, our proposed CNN model is also relatively shallow to avoid overfitting. The CNN architecture can be found in Table I. In the CNN model, we use two different activation functions, sigmoid following Equation (2) and SiLU following Equation (3).

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

$$\text{SiLU}(x) = x \cdot \text{sigmod}(x) = \frac{x}{1 + e^{-x}} \tag{3}$$

The sigmoid function in the last layer of the CNN model maps all values into the range of $[0, 1)$, which indicates the likelihood that an active target user exists in the input segment. Therefore, the prediction module outputs $N = 43$ likelihoods for each 1-second capture. These likelihoods are then sent to the decision module for the final detection result.

C. Decision Module

The decision module takes the likelihood scores of all segments in a single capture as input and outputs a single binary detection result. As shown in Figure 3, we first perform thresholding on all likelihood scores following Equation (4), where p_k is the likelihood score of the $k-$th segment and $k \in \{1, 2, \ldots, N\}$.

$$d_k = \mathbb{1}(p_k \geq \tau) \tag{4}$$

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TABLE II: Reproducibility checklist.

Item	Specification
CPU	AMD Threadripper 5975WX
GPU	Nvidia 3090
Python	3.10
NumPy	1.26
Tensorflow	2.15

TABLE III: CNN training details.

	1st round	2nd round	3rd round
α_{init}	10^{-4}	10^{-5}	6×10^{-5}
α_{min}	5×10^{-8}	5×10^{-8}	5×10^{-8}
β	0.5	0.1	0.1
γ_{decay}	10	15	15
γ_{stop}	25	50	50
Epochs	100	500	500
Freeze	No	No	Yes
Batch size	4		
Optimizer	AdamW [17]		

After thresholding, we obtain N initial detection results $d_k \in \{0, 1\}$. Because a target user can be considered as "existing" regardless of how long it is active in the 1-second capture, we don't need $d_k = 1$ for all segments to confirm the detection of an active target user. Therefore, we have a voting block at the end following Equation 5, where o is the final binary decision for a 1-second capture and v is the minimum required number of positive segments in the capture.

$$o = \mathbb{1}(\sum_{k=1}^{N} d_k \geq v) \tag{5}$$

Here, we experimentally find $\tau = 0.92$ and $v = 2$. The details of finding these parameters are discussed in Section IV-D.

D. Implementation Details

In this section, we discuss the implementation details of the SCD framework, including how the CNN model is trained and how the decision module parameters are selected. To ensure the reproducibility of our framework, we list the hardware and software requirements in Table II.

The CNN model is trained in 3 different rounds. In every round, we have a learning rate decay schedule. Specifically, we have an initial learning rate α_{init} and a minimum learning rate α_{min}. The learning rate is decayed by a factor of β when the validation loss has not improved for γ_{decay} epochs. Finally, the model stops training when validation loss has not improved for γ_{stop} epochs and the model weights with the lowest validation loss is saved. The necessary training details are listed in Table III, where the freeze means freezing the parameters in the first two convolutional layers and the first two batch normalization layers in the CNN model.

In the first training round, we want the model to learn to extract features which can roughly distinguish between the positive and negative captures. Therefore, we only use the data from the easy category in this round. Then, in the second training round, we fine-tune the model so that it can learn to extract additional subtle features distinguishing between the hard examples. Therefore, we decrease the learning rate and include the captures from the hard category in this round. Finally, in the third training round, we further improve the decision boundary for the hard examples. Therefore, we freeze the model parameters in the early layers. At the same time, we add a few more hard captures to increase the diversity of the training data. As mentioned in Section III, we perform the 64-16-20 split on both categories. It should be noted that all segments from the same capture are put into the same set to avoid overfitting. The details of how the data is used in each round are listed in Table IV.

After the model is trained, we apply it on all 240 captures and obtain the 43 likelihoods for each capture, which are used to determine the parameters τ and v in the decision module. We rely on the following heuristics to determine the parameters. First, the target user does not need to be active for the entire 1-second capture. As a result, we tend to select a small v. On the other hand, we have more positive training samples (160 captures with active user) than negative samples (80 captures without active user). Therefore, the trained CNN model should be naturally biased towards positive results. As a result, we tend to select a large τ to counteract this bias. Following these two heuristics, we experimentally find the combination of $\tau = 0.92$ and $v = 2$ can allow most true positives while eliminating any false positives across all 240 captures.

E. Computational Complexity and Efficiency

In this section, we discuss the computational complexity and efficiency of our proposed SCD in inference stage. As shown in Figure 3, the bottleneck of both space and time complexity exists before the downsampling step in the pre-processing module. Specifically, the space complexity is bottlenecked by the segmenting step, which has a memory requirement of $O(NT)$. At the same time, the time complexity is bottlenecked by the FFT operation, which requires a time of $O(T \log T)$. It should be noted that all segments should be able to be processed in parallel so that the time is independent of N.

At the same time, we can trade off between the space and time complexity. Specifically, if we perform the FFT operation all the way through downsampling for each single segment every time, the space complexity could be reduced to $O(M)$ where M is the number of samples from the input capture. However, we increase the time complexity because we cannot perform parallel computation across different segments. Therefore, the resulted time complexity will be $O(NT \log T)$ instead. In summary, we have a space complexity of $O(NT)$ and time complexity of $O(T \log T)$ in the current implementation, which can be modified to obtain a space complexity of $O(M)$ and time complexity of $O(NT \log T)$ according to different space and computational complexity requirements.

In our development environment as specified in Table II, the average runtime from loading a numpy file of 10 million complex-valued I/Q samples until returning a binary output is 0.75 second.

V. Experiments

In the experiments, we use precision and recall as the metrics. The definitions are shown in Equations (6) and (7), where TP

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TABLE IV: CNN training data split.

	1st round		2nd round		3rd round	
	Train	Validation	Train	Validation	Train	Validation
64% in easy	✓		✓		✓	
16% in easy		✓		✓		✓
64% in hard			✓		✓	
16% in hard				✓	✓	
20 % in hard						✓

TABLE V: Experimental results on all available training captures.

Experiment	Precision	Recall	Accuracy
Without segmenting	100%	87.5%	91.6%
Without decision ($v = 1$)	87.4%	99.4%	90.0%
Without decision ($v = N$)	100%	84.4%	93.8%
1-round training	100%	87.5%	91.6%
2-round training	100%	96.3%	97.5%
SCD (proposed)	100%	97.5%	98.3%

means true positive, FP is false positive and FN is false negative.

$$\text{Precision} = \frac{\text{TP}}{\text{TP + FP}} \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP + FN}} \tag{7}$$

We perform ablation studies to show that all three rounds of training are able to improve the overall performance. Besides, we show that the segmenting in the pre-processing stage and the final decision module with thresholding and voting are also necessary. In the experiment without segmenting, we still select a threshold to eliminate any FP following the second heuristic discussed in Section IV-D. In the experiments without decision module, we test $v = 1$ and $v = N$ with $\tau = 0.5$ in both cases. The results of different experiments across all 240 available captures are listed in Table V.

As shown in Table V, we can observe that the proposed SCD framework benefits from every round of training as well as the segmenting and decision module. At the same time, the final SCD model achieves 89% accuracy on the hidden test set with 95% precision and 82% recall.

VI. Conclusion

In this work, we consider the problem of signal detection in wideband spectrum data with realistic channel conditions and unknown interferences. To address this problem, we propose the SCD framework, which is developed on real spectrum data to detect active transmissions of 5G-like signals. SCD is able to achieve 98.3% accuracy across all available training captures and 89% accuracy across the captures in a hidden testset. In addition, SCD is flexible in the tradeoff between the space and time complexity if a specific computational resource is limited.

It should be noted that the parameters in the current SCD framework are selected based on the available training captures. In fact, our proposed framework and development procedures can be easily adapted for different capture durations, spectrum bandwidth or target use transmissions by selecting a separate list of parameters and retraining the CNN accordingly.

In the future, we will try to improve the performance of SCD by appropriate data augmentation and more exhaustive and systematic selection of the parameters. At the same time, we will also test SCD in different signal detection tasks.

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